

Physics-Integrated Generative Edge-AI Framework for Explainable Aero-Engine Prognostics

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Abstract—Prognostics and Health Management of aircraft turbofan engines is a critical challenge in aerospace engineering, where accurate Remaining Useful Life (RUL) prediction enables proactive maintenance, reduces unplanned downtime, and improves operational safety. Existing approaches, from recurrent networks to attention-based transformers, predominantly deliver single-point RUL estimates with no uncertainty quantification and no mechanism to verify whether predictions respect the physical laws governing engine degradation. This work proposes a Physics-Integrated Generative Edge-AI Framework trained on the NASA C-MAPSS FD001 benchmark that replaces scalar point prediction with generative counterfactual prognostics. An attention-based encoder produces latent degradation embeddings that condition a physics-constrained conditional variational autoencoder (cVAE) to sample multiple plausible future degradation trajectories. Physics-derived constraints including degradation monotonicity and thermodynamic consistency are embedded directly into the training loss rather than applied post-hoc. A dual-head output delivers both a probabilistic RUL distribution and a physics-violation risk score per trajectory. A post-prediction Reliability Index feeds a three-tier gating layer producing ACCEPT, WARN, or REJECT decisions. Explainability is quantified through the novel Counterfactual Physical Consistency (CPC) metric, measuring whether generated trajectories remain physically coherent under shifted operating conditions. The system is deployed via a Streamlit dashboard, FastAPI backend, and an MQTT-connected live digital twin.

Index Terms—remaining useful life, prognostics, physics-informed learning, generative model, digital twin, C-MAPSS, reliability gating, counterfactual

I. INTRODUCTION

[Placeholder — introduction to be added.]

II. RELATED WORK

The following works are reviewed with emphasis on existing implementations and the key research gaps this work addresses.

Classical and statistical methods. Early RUL estimators relied on physics-based wear models and autoregressive time-series approaches. Saxena et al. [1] established the C-MAPSS benchmark and its standard evaluation metrics, RMSE and the asymmetric NASA score function, which remain the community reference. A key limitation is that the thermodynamic physics embedded in the simulator are not reused as training constraints in downstream models, and no trajectory-level ground truth exists for future degradation paths.

Ensemble and gradient-boosted methods. Random Forest and LightGBM pipelines demonstrated competitive FD001

performance through hand-crafted degradation features [5]. These are purely discriminative models with no uncertainty quantification or trajectory modelling capability.

Recurrent deep learning. LSTM and GRU networks removed the need for manual feature engineering. Heimes [2] and Zheng et al. [3] showed that sequence models and convolutional windows over sensor data yield strong results on C-MAPSS. All outputs are deterministic scalars with no mechanism to flag physically implausible predictions.

Attention mechanisms. Zhang et al. [4] demonstrated that dual-aspect self-attention applied across temporal and sensor-wise dimensions outperforms vanilla LSTMs on C-MAPSS. Despite this improvement, the output remains a single scalar, explainability relies on post-hoc attribution such as SHAP, and physics are absent from the model.

Physics-informed neural networks. PINNs embed domain knowledge such as degradation monotonicity and thermodynamic constraints into training objectives, improving generalisation under data-scarce conditions. Most existing works, however, apply physics as post-hoc constraints or input features rather than embedding them into a generative model's training loss.

Uncertainty quantification and digital twins. Bayesian dropout and conformal prediction approaches have introduced statistical uncertainty bounds into RUL predictions [7]. MQTT-based streaming pipelines with live virtual replicas represent the current industrial frontier for real-time PHM [8]. The remaining gap is that Bayesian uncertainty is purely statistical: no existing approach supports physically coherent trajectory generation or counterfactual reasoning over RUL.

Summary of gaps. The literature converges on three unresolved problems this work addresses: (1) all mainstream models produce point estimates with no trajectory-level uncertainty; (2) physics knowledge is applied post-hoc rather than embedded in the generative process; and (3) no existing work supports physically grounded counterfactual reasoning over RUL under shifted operating conditions.

III. PROPOSED METHODOLOGY

The proposed framework processes multivariate sensor streams from either a CSV batch source or a live MQTT digital twin, encodes them into latent degradation embeddings, feeds those into a physics-constrained cVAE to generate trajectory distributions, and surfaces results through a dual-head output

followed by reliability gating. Stages 1 and 2 are shared across all model modes. Gating (stage 6) is strictly post-prediction; raw model output is always preserved for analysis. The end-to-end pipeline is described in Table I.

TABLE I
END-TO-END GENERATIVE RUL PREDICTION PIPELINE

Stage	Description
① Data ingestion	CSV batch upload or live MQTT stream; 21 sensors + 3 operational settings per cycle.
② Preprocessing	Min-max normalisation; 30-cycle sliding window; near-zero variance sensor removal; shared across all model modes.
③ Attention encoder	Multivariate sensor window encoded into a latent degradation embedding via self-attention (temporal and sensor-wise).
④ cVAE generative model	Physics constraints (degradation monotonicity, thermodynamic bounds) embedded in training loss; samples N plausible future degradation trajectories conditioned on the latent embedding.
⑤ Dual-head output	<i>Head A</i> : probabilistic RUL distribution (mean, variance, CI). <i>Head B</i> : physics-violation risk score per trajectory. Raw prediction always preserved.
⑥ Reliability gating	Reliability Index quantifies per-engine prediction trustworthiness; CPC metric computed; assigns ACCEPT, WARN, or REJECT.
⑦ Output surfaces	Streamlit dashboard; FastAPI REST API; headless CLI; TFLite edge export.

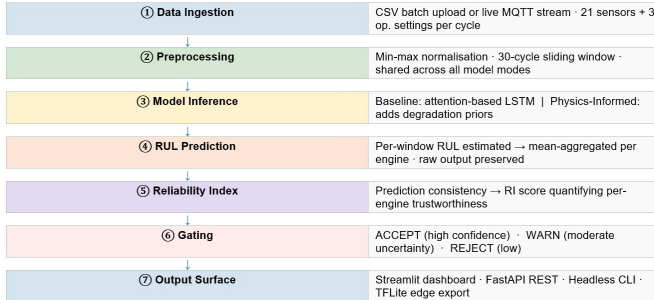


Fig. 1. End-to-end generative RUL prediction pipeline with physics constraints and post-prediction reliability gating.

IV. DATASET AND EXPERIMENT PLAN

A. Primary Dataset: NASA C-MAPSS FD001

The NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset is used as the primary benchmark [1]. FD001 consists of a single operating condition and single fault mode (HPC degradation), with 100 training and 100 test engine run-to-failure trajectories. Each trace provides 21 sensor channels and 3 operational settings per time cycle. FD001 is selected for Phase 1 due to its controlled structure, which isolates validation of the generative and physics-constraint mechanisms from multi-regime complexity. Table II summarises all four subsets and their planned use.

TABLE II
NASA C-MAPSS SUBSETS AND PROJECT USAGE ALLOCATION

Subset	Train	Test	Sens.	Op.	Project Use
FD001 (primary)	100	100	21	1	Training & evaluation
FD002	260	259	21	6	Multi-condition test
FD003	100	100	21	1	Multi-fault eval.
FD004	248	249	21	6	Full complexity

B. Preprocessing Protocol

- **RUL labelling:** Piecewise linear with 125-cycle cap, reflecting the stable early-life operating phase.
- **Normalisation:** Min-max per feature computed on the training set; same parameters applied to the test set.
- **Windowing:** 30-cycle sliding window; early-cycle sequences zero-padded at the front.
- **Sensor selection:** Near-zero variance sensors ($s_1, s_5, s_6, s_{10}, s_{16}, s_{18}, s_{19}$) excluded, retaining 14 informative channels.

C. Evaluation Metrics and Validation

- **RMSE:** Symmetric deviation penalty on the FD001 test set; enables direct comparison with deterministic baselines.
- **NASA score function:** Asymmetric penalty, heavier for late predictions than early ones; reflects real-world maintenance risk asymmetry.
- **RI acceptance rate:** Proportion of test engines receiving ACCEPT gate under calibrated thresholds.
- **Regression tests:** Baselines frozen at first validation run; drift checked via `test_inference_regression.py`.
- **LightGBM backend:** Benchmarked via `benchmark_model_backends.py` before any promotion.

V. RESULTS AND DISCUSSION

[Placeholder — to be completed in later milestones.]

VI. CONCLUSION

[Placeholder — to be completed in later milestones.]

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